

Gesture Classification from Surface EMG Signals

Synapse Problem Statement

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1 Introduction

Surface electromyography (sEMG) signals capture the electrical activity generated by motor unit action potentials during voluntary muscle contraction. Although sEMG provides rich neuromuscular information, it is inherently non-stationary, noisy, and strongly subject-dependent, making robust gesture classification challenging.

This work presents an end-to-end sEMG gesture classification pipeline that integrates physiologically motivated signal preprocessing, interpretable feature engineering, and subject-aware machine learning. The system is explicitly designed to avoid data leakage, generalize across subjects, and be evaluated using strict, class-balanced metrics.

2 Dataset Description

The dataset consists of raw sEMG recordings from 25 subjects performing five distinct hand gestures across multiple sessions. Each recording contains synchronized signals from eight sEMG channels placed circumferentially around the forearm to capture spatial muscle activation patterns.

Each CSV file corresponds to a single trial of a specific gesture. Subject and session metadata are preserved and explicitly used to enforce subject-independent evaluation.

3 Preprocessing Pipeline

3.1 DC Offset Removal and Bandpass Filtering

Raw sEMG signals contain DC offsets from electrode-skin interfaces, low-frequency motion artifacts, and high-frequency electronic noise. Since physiologically meaningful sEMG activity primarily lies within 20–250 Hz, spectral constraints are imposed.

Given a sampling frequency of 512 Hz, a 4th-order Butterworth bandpass filter with cutoffs at 20 Hz and 250 Hz is applied. Zero-phase forward-backward filtering is used to preserve temporal structure, which is critical for downstream feature extraction.

3.2 Rectification and RMS Normalization

Filtered signals are full-wave rectified to convert bipolar oscillations into non-negative activation magnitudes, consistent with the physiological interpretation of muscle excitation.

RMS-based normalization is applied independently per trial and channel to reduce inter-subject variability arising from muscle strength, electrode placement, and skin impedance differences. RMS provides a stable estimate of average muscle activation power.

Importantly, both **normalized** and **non-normalized** signal representations are retained. Normalized signals emphasize relative activation patterns and inter-channel coordination that generalize across subjects, while non-normalized signals preserve absolute amplitude and spectral power information related to force and muscle recruitment. Different feature groups intentionally operate on different representations.

3.3 Windowing Strategy

Continuous recordings are segmented into overlapping temporal windows using a fixed window size and stride. This converts the non-stationary signal into locally quasi-stationary segments, enabling stable feature computation while preserving gesture dynamics.

4 Feature Extraction

Feature extraction is designed to capture complementary physiological, spectral, spatial, and temporal characteristics of muscle activation.

4.1 Time-Domain Features (Normalized Signals)

For each channel:

- **RMS (normalized):** Encodes relative activation strength independent of subject scaling.
- **Mean Absolute Value (MAV):** Represents sustained contraction level and correlates with muscle effort.
- **Waveform Length (WL):** Captures cumulative signal variation, reflecting both amplitude modulation and firing complexity.
- **Slope Sign Changes (SSC):** Quantifies rapid activation changes indicative of fine motor control.

4.2 Frequency-Domain Features (Non-Normalized Signals)

Using Welch’s power spectral density estimate:

- **Zero Crossing Rate (ZC):** Reflects dominant frequency content and firing variability.
- **Mean Frequency (MNF):** Represents the spectral centroid, sensitive to contraction dynamics and fatigue.
- **Median Frequency (MDF):** Splits spectral energy into equal halves, providing robustness to localized spectral peaks.

Non-normalized spectral features preserve physiologically meaningful amplitude information that would be lost under normalization.

4.3 Inter-Channel Features (Normalized Signals)

- **IQR Spread of Channel RMS Values:** Measures dispersion in muscle activation using interquartile range, which is robust to outlier channels caused by transient noise or electrode artifacts.
- **Dominance Ratio (90th percentile / median):** Quantifies whether a gesture is driven by a small subset of muscles, using median-based normalization to avoid skew.
- **Entropy of Channel Energy Distribution:** Captures how evenly activation is distributed across muscles, reflecting coordination versus localization.
- **Mean and Standard Deviation of Pairwise Correlations:** Encode overall muscle co-activation and variability in synergy.
- **Maximum Absolute Inter-Channel Correlation:** Identifies strongly synchronized muscle pairs characteristic of specific gestures.

4.4 Hjorth Parameters

For each channel:

- **Mobility:** Estimates dominant frequency relative to signal variance.
- **Complexity:** Measures deviation from simple harmonic motion, reflecting neuromuscular control complexity.

4.5 Temporal Evolution Features

To capture gesture-level dynamics beyond isolated windows, temporal features are computed once per trial and broadcast across windows:

- **RMS Slope:** Gradual activation trends.
- **RMS Volatility:** Variability in activation across windows.
- **Net RMS Change:** Overall activation transition during gesture execution.

5 Model Selection and Hyperparameter Optimization

Multiple classifiers were evaluated using GridSearchCV with subject-wise Group K-Fold cross-validation and macro-averaged F1 score as the optimization objective.

The following model families were explored: Logistic Regression, k-Nearest Neighbors, Linear Support Vector Machines, Random Forests, and Gradient Boosting. Tree-based ensemble methods consistently outperformed linear and distance-based models, indicating the importance of non-linear feature interactions. Gradient Boosting achieved the strongest window-level performance and was selected for the final pipeline.

6 Model Training and Evaluation

Window-level classification yielded macro F1 scores in the range of 0.45–0.50 across models, reflecting the inherent ambiguity of short sEMG windows.

To exploit temporal consistency within trials, window-level predictions were aggregated using majority voting. This significantly improved robustness, resulting in trial-level macro F1 scores of **0.71–0.72**. Macro F1 ensures balanced performance across gesture classes and prevents dominance by easily separable gestures.

7 Inference Pipeline

Inference mirrors the training pipeline exactly. Window-level features are extracted, scaled using training statistics, classified using the trained model, and aggregated via majority voting to produce a single gesture label per trial.

8 Discussion and Future Work

Future work includes feature ablation studies, subject-adaptive normalization, and exploration of lightweight neural architectures. The current pipeline provides a strong, interpretable baseline suitable for real-world sEMG gesture recognition.

9 Conclusion

We present a physiologically grounded and interpretable sEMG gesture classification system that combines principled signal processing with robust machine learning evaluation. Temporal aggregation enables strong trial-level performance while maintaining transparency, reproducibility, and generalization across subjects.